CHAPTER 11

INTRODUCTION TO THREE DIMENSIONAL GEOMETRY

CHAPTER ESSENCE AT A GLANCE:

Welcome to the 3rd dimension! Adding the Z-axis extends plane geometry into full 3D space. The 3 coordinate planes (XY, YZ, ZX) divide space into 8 Octants. The distance formula simply gets an extra $(z_2 - z_1)^2$ term inside the square root!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **3D Coordinate Axes & Octants:** 3 mutually perpendicular axes divide space into 8 Octants. Points on X-axis have (x, 0, 0); points in XY-plane have (x, y, 0).

• **3D Distance Formula:** Distance between P(x_1, y_1, z_1) and Q(x_2, y_2, z_2) is d = sqrt($(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$). Distance from origin OP = sqrt($x^2 + y^2 + z^2$).

• **Section Formula (Internal Division):** P divides AB in ratio m:n: P = [(m x_2 + n x_1) over (m+n), (m y_2 + n y_1) over (m+n), (m z_2 + n z_1) over (m+n)].

• **Midpoint & Centroid:** Midpoint: [(x_1+x_2) over 2, (y_1+y_2) over 2, (z_1+z_2) over 2]. Triangle Centroid: [(x_1+x_2+x_3) over 3, (y_1+y_2+y_3) over 3, (z_1+z_2+z_3) over 3].

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Euclidean Distance Formula in Three-Dimensional Space

$$P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n}\right)$$

Section Formula for Internal Division in Ratio m : n

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}\right)$$

Centroid Coordinates of Spatial Triangle

$$\text{Octants: } (\pm x, \pm y, \pm z) \rightarrow 8 \text{ Disjoint Spatial Quadrant Compartments}$$

Eight Octant Spatial Sign Symmetry Matrix

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** To find coordinates of a point on the X-axis, set y = 0 and z = 0: (x, 0, 0). On the XY-plane, only z = 0: (x, y, 0).

• **Alert:** The 3D distance from origin (0,0,0) to point P(x, y, z) is simply OP = sqrt($x^2 + y^2 + z^2$)!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Find the distance between the points P(1, -3, 4) and Q(-4, 1, 2). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Distance Formula in 3D]** d = $\sqrt{((x2 - x1)^2 + (y2 - y1)^2 + (z2 - z1)^2)}$. [1 Mark]

• **[Step 2: Substitution & Evaluation]** d = $\sqrt{((-4 - 1)^2 + (1 - (-3))^2 + (2 - 4)^2)} = \sqrt{((-5)^2 + 4^2 + (-2)^2)} = \sqrt{(25 + 16 + 4)} = \sqrt{(45)} = 3\sqrt{5}$ units. [1 Mark]

• **[Boxed Result: Distance = $3\sqrt{5}$ units $\approx$ 6.71 units]**

2. Find the coordinates of the midpoint of the line segment joining the points P(2, 3, 5) and Q(4, 3, 1). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Midpoint Formula]** M = ((x1 + x2) over 2, (y1 + y2) over 2, (z1 + z2) over 2). [1 Mark]

• **[Step 2: Calculation]** M = ((2 + 4) over 2, (3 + 3) over 2, (5 + 1) over 2) = (6 over 2, 6 over 2, 6 over 2) = (3, 3, 3). [1 Mark]

• **[Boxed Result: Midpoint = (3, 3, 3)]**

3. Find the coordinates of the point which divides the line segment joining the points (-2, 3, 5) and (1, -4, 6) in the ratio 2 : 3 internally. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Section Formula]** P = ((m x2 + n x1) over (m + n), (m y2 + n y1) over (m + n), (m z2 + n z1) over (m + n)). Here m = 2, n = 3. [1 Mark]

• **[Step 2: Coordinate Substitutions]** x = [2(1) + 3(-2)] over (2 + 3) = (2 - 6) over 5 = -4 over 5. y = [2(-4) + 3(3)] over 5 = (-8 + 9) over 5 = 1 over 5. z = [2(6) + 3(5)] over 5 = (12 + 15) over 5 = 27 over 5. [1.5 Marks]

• **[Step 3: Point Coordinates]** P = (-4 over 5, 1 over 5, 27 over 5). [0.5 Mark]

• **[Boxed Result: Point = (-4 over 5, 1 over 5, 27 over 5)]**

4. Find the coordinates of the point which divides the line segment joining the points P(1, -2, 3) and Q(3, 4, -5) in the ratio 2 : 3 externally. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: External Section Formula]** P = ((m x2 - n x1) over (m - n), (m y2 - n y1) over (m - n), (m z2 - n z1) over (m - n)). Here m = 2, n = 3, m - n = -1. [1 Mark]

• **[Step 2: Coordinate Calculations]** x = [2(3) - 3(1)] over (-1) = (6 - 3) over (-1) = -3. y = [2(4) - 3(-2)] over (-1) = (8 + 6) over (-1) = -14. z = [2(-5) - 3(3)] over (-1) = (-10 - 9) over (-1) = 19. [1.5 Marks]

• **[Step 3: Coordinates]** External division point = (-3, -14, 19). [0.5 Mark]

• **[Boxed Result: Point = (-3, -14, 19)]**

5. Show that the points P(-2, 3, 5), Q(1, 2, 3), and R(7, 0, -1) are collinear. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Distance PQ]** PQ = $\sqrt{((1 - (-2))^2 + (2 - 3)^2 + (3 - 5)^2)} = \sqrt{(3^2 + (-1)^2 + (-2)^2)} = \sqrt{(9 + 1 + 4)} = \sqrt{(14)}$. [1 Mark]

• **[Step 2: Distance QR and PR]** QR = $\sqrt{((7 - 1)^2 + (0 - 2)^2 + (-1 - 3)^2)} = \sqrt{(6^2 + (-2)^2 + (-4)^2)} = \sqrt{(36 + 4 + 16)} = \sqrt{(56)} = 2\sqrt{(14)}$. PR = $\sqrt{((7 - (-2))^2 + (0 - 3)^2 + (-1 - 5)^2)} = \sqrt{(9^2 + (-3)^2 + (-6)^2)} = \sqrt{(81 + 9 + 36)} = \sqrt{(126)} = 3\sqrt{(14)}$. [1.5 Marks]

• **[Step 3: Collinearity Condition]** PQ + QR = $\sqrt{(14)} + 2\sqrt{(14)} = 3\sqrt{(14)}$ = PR. Since PQ + QR = PR, the points P, Q, and R are collinear. [0.5 Mark]

• **[Boxed Result: PQ + QR = PR = $3\sqrt{(14)}$; Points are collinear]**

6. Find the equation of the set of points P such that its distances from the points A(3, 4, 5) and B(-1, 3, -7) are equal. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Condition PA = PB]** $PA^2 = PB^2$. Let P be (x, y, z). [1 Mark]

• **[Step 2: Distance Equations]** $(x - 3)^2 + (y - 4)^2 + (z - 5)^2 = (x + 1)^2 + (y - 3)^2 + (z + 7)^2$. $x^2 - 6x + 9 + y^2 - 8y + 16 + z^2 - 10z + 25 = x^2 + 2x + 1 + y^2 - 6y + 9 + z^2 + 14z + 49$. [1.5 Marks]

• **[Step 3: Simplification]** $-6x - 8y - 10z + 50 = 2x - 6y + 14z + 59$. $-8x - 2y - 24z - 9 = 0 \Rightarrow 8x + 2y + 24z + 9 = 0$. [1.5 Marks]

• **[Boxed Result: 8x + 2y + 24z + 9 = 0]**

7. Find the coordinates of the points which trisect the line segment joining the points P(4, 2, -6) and Q(10, -16, 6). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: First Trisection Point A (1 : 2)]** Divides PQ internally in ratio 1 : 2. A = ([1(10) + 2(4)] over 3, [1(-16) + 2(2)] over 3, [1(6) + 2(-6)] over 3) = (18 over 3, -12 over 3, -6 over 3) = (6, -4, -2). [2 Marks]

• **[Step 2: Second Trisection Point B (2 : 1)]** Divides PQ internally in ratio 2 : 1 (or midpoint of A and Q). B = ([2(10) + 1(4)] over 3, [2(-16) + 1(2)] over 3, [2(6) + 1(-6)] over 3) = (24 over 3, -30 over 3, 6 over 3) = (8, -10, 2). [2 Marks]

• **[Boxed Result: Trisection points are (6, -4, -2) and (8, -10, 2)]**